

Figure 6.5 A JWE in Compact Serialization consists of 5 components: a header, an encrypted key (blank in this case), an Initialization Vector or nonce, the encrypted ciphertext, and then the authentication tag. Each component is URL-safe Base64-encoded. Values have been truncated for display.

JWE divides specification of the encryption algorithm into two parts:

- The `enc` header describes the authenticated encryption algorithm used to encrypt the payload of the JWE.
- The `alg` header describes how the sender and recipient agree on the key used to encrypt the content.

There are a wide variety of key management algorithms for JWE, but for this chapter you will stick to direct encryption with a secret key. For direct encryption the algorithm header is set to `"dir"` (direct). There are currently two available families of encryption methods in JOSE, both of which provide authenticated encryption:

- `A128GCM`, `A192GCM`, and `A256GCM` use AES in *Galois Counter Mode* (GCM).
- `A128CBC-HS256`, `A192CBC-HS384`, and `A256CBC-HS512` use AES in *Cipher Block Chaining* (CBC) mode together with either